Government Spending Efficiency Analysis

Health Sector - County-Level Evaluation in Kenya

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1. Executive Summary

This study evaluates how efficiently counties in Kenya convert public health expenditure into measurable healthcare outcomes. Using county-level data on budgets, population, and mortality indicators, an efficiency scoring model was developed to compare performance across counties.

The analysis reveals that:

- Higher spending does not consistently translate into better health outcomes.
- Some counties achieve strong outcomes with relatively low spending, indicating high efficiency.
- Others exhibit high spending but poor outcomes, suggesting inefficiencies in resource utilization.

The findings highlight significant disparities in efficiency and provide a basis for targeted policy interventions, including improved allocation strategies and adoption of best practices from high-performing counties.

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2. Objectives

The primary objective of this study is to:

- Measure the efficiency of county-level health spending
- Compare counties based on outcomes relative to expenditure
- Identify high-performing and underperforming counties
- Provide data-driven recommendations for improving efficiency

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3. Data and Methodology

3.1 Data Sources

The analysis integrates multiple datasets, including:

- County health budget allocations and budget breakdown (recurrent vs development) (2019)
- Population data (2019 census)
- Mortality indicators (2019):
    - Infant mortality rate
    - Maternal mortality rate
    - Under-5 mortality rate
- Hospital Maternal, neonatal, and foetal deaths (2016–2020)

All datasets were standardized and aligned to a common reference period (primarily 2019, with some use of 2018–2020 averages).

3.2 Data Processing Framework

A Medallion Architecture was used:

- **Bronze Layer:** Raw data ingestion
- **Silver Layer:** Data cleaning, normalization, and feature engineering
- **Gold Layer:** Analytical dataset and efficiency scoring

Key derived metrics include:

- Health spending per capita
- Mortality-based outcome scores
- Composite efficiency score

3.3 Efficiency Model

Efficiency is defined as:

Ability to achieve better health outcomes with lower spending

Steps:

1. Normalize all variables (0–1 scale)
2. Invert mortality rates (since lower is better)

3. Compute an **Outcome Score** (average of mortality indicators)
4. Adjust by spending to obtain **Cost Efficiency**
5. Scale to a final **Efficiency Score (0–100)**

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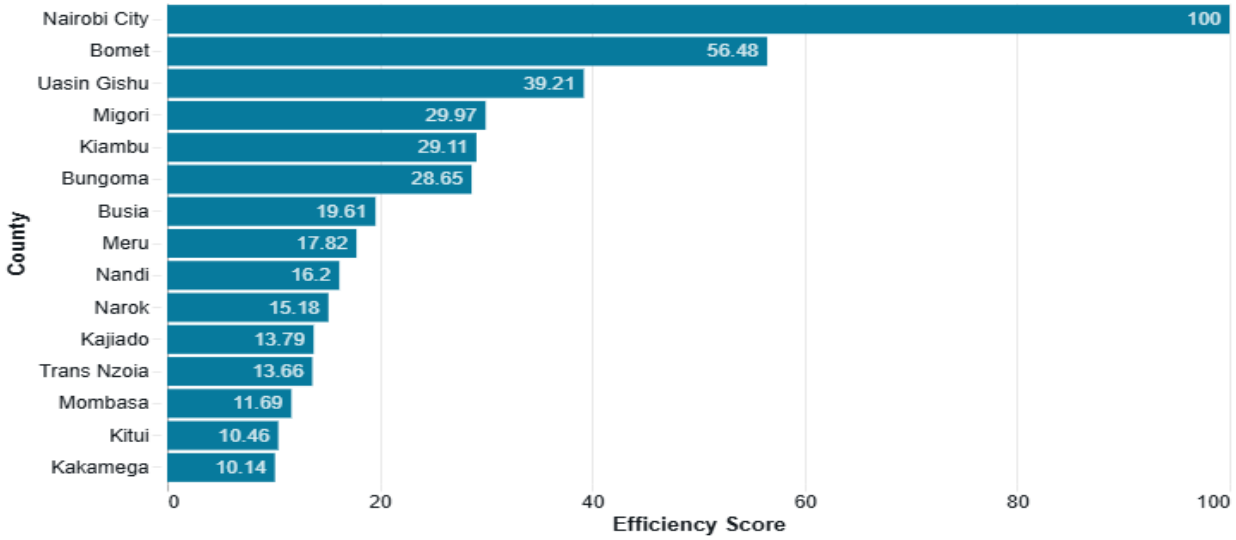
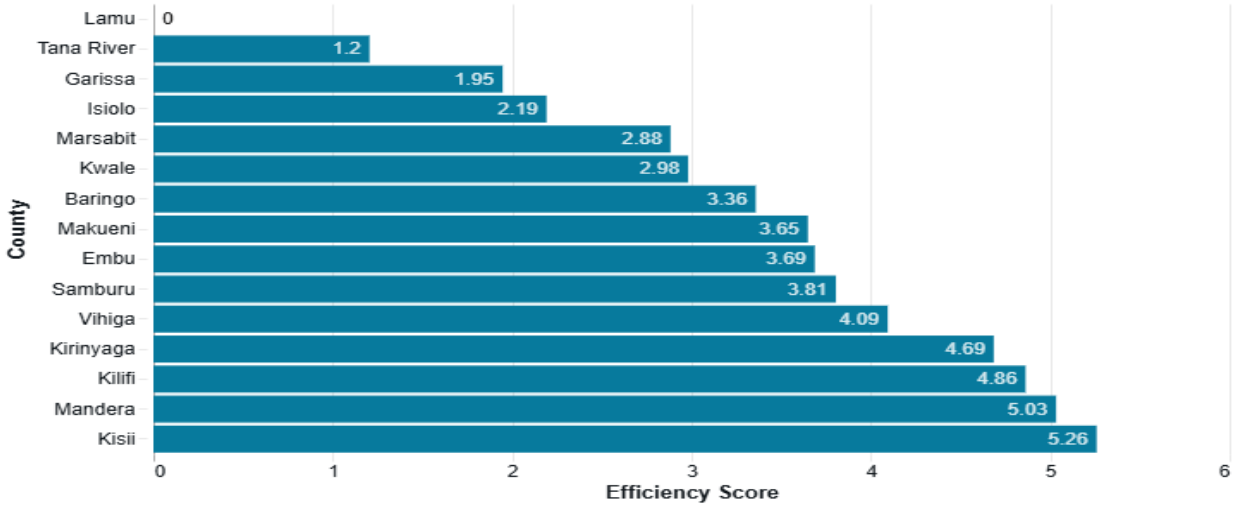
4. Key Findings

4.1 County Efficiency Rankings

Top 10 Efficient Counties



Bottom 10 Counties



There are clear leaders and laggards in efficiency.

Insights:

- Top-performing counties achieve strong outcomes with relatively low spending
- Bottom-performing counties often combine high spending with poor outcomes

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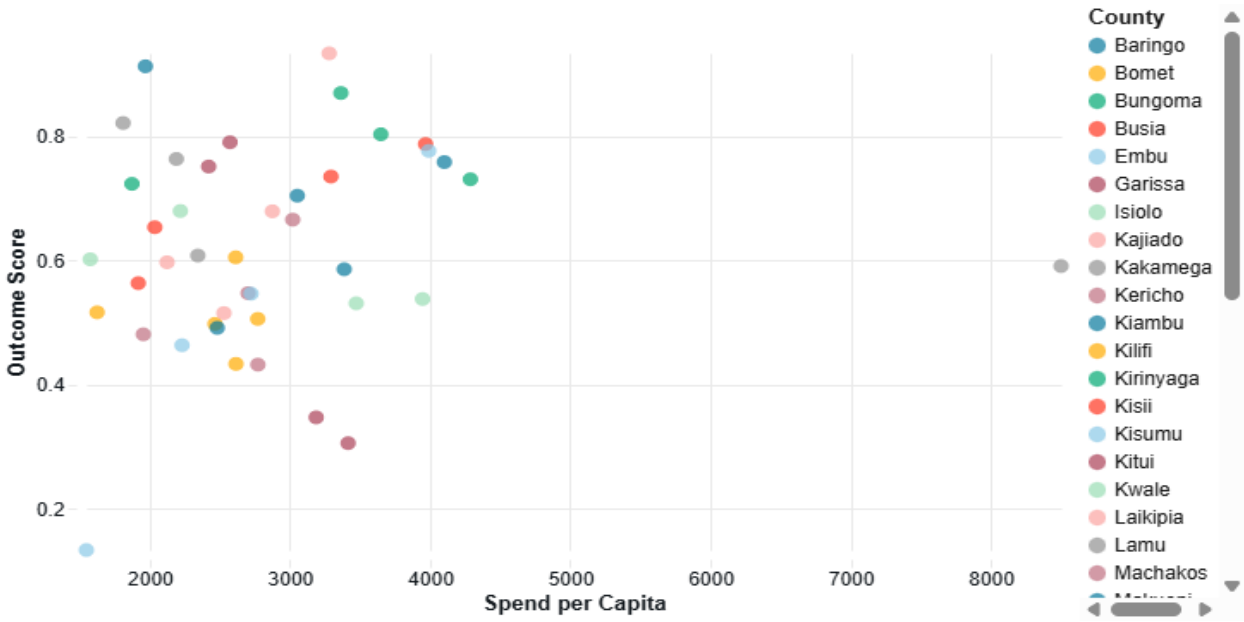
4.2 Spending vs Health Outcomes

Spend per Capita vs Outcome Score



This visualization examines the relationship between spending and outcomes.

Insights:

- Weak or inconsistent correlation between spending and outcomes
- Presence of:
    - High-spend, low-outcome counties (bottom-right) - Inefficient
    - Low-spend, high-outcome counties (top-left) - Efficient

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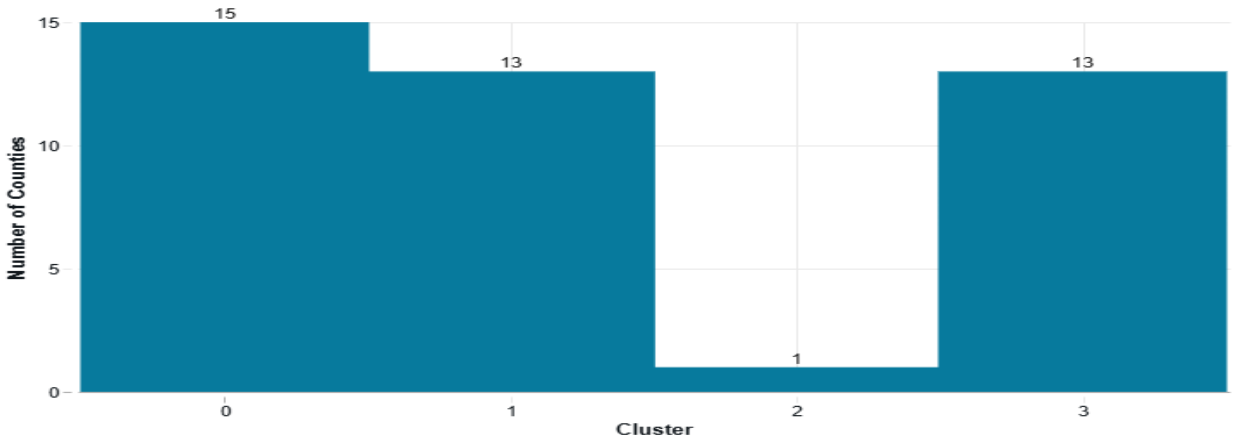
4.4 County Clusters (Performance Segmentation)

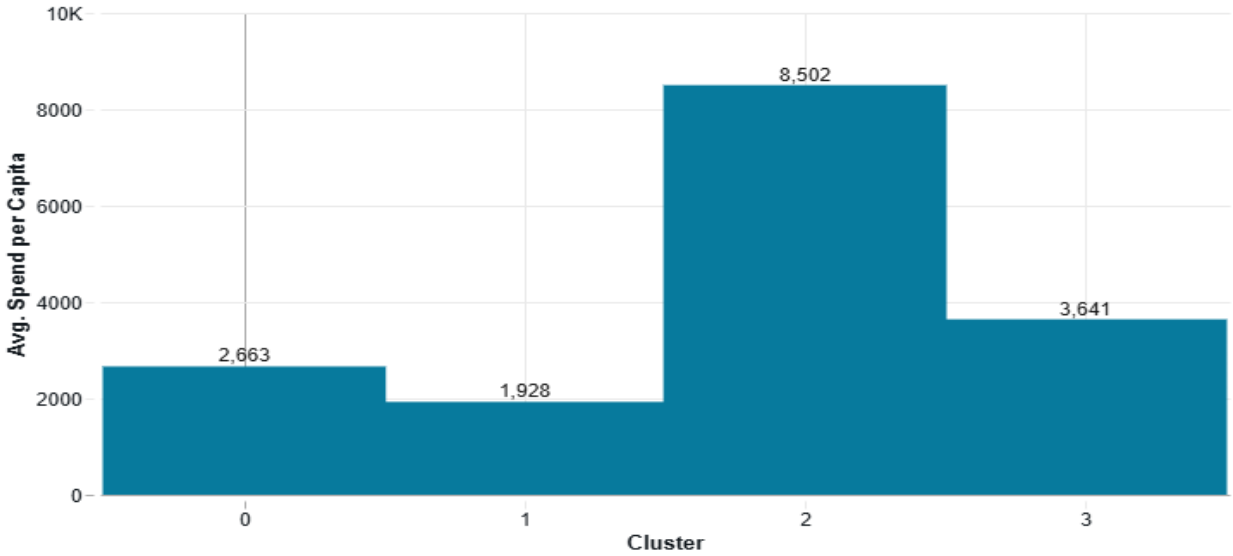
Counties were grouped into clusters based on spending and outcomes.

Key segments identified:

- **High Efficiency:** Moderate to high spending, strong outcomes (Cluster 3)
- **Moderate Efficiency:** Low spending, acceptable outcomes (Cluster 1)
- **Inefficient:** Moderate spending, weak outcomes (Cluster 0)
- **High-cost inefficient:** High spending, poor outcomes (Cluster 2)

Cluster Distribution





This segmentation provides a clearer framework for targeted interventions.

4.5 Health Budget Allocation vs Outcomes

Health Share vs Outcome Score

This analysis evaluates whether prioritizing health in budgets improves outcomes.

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• Weak correlation - inefficiency in allocation
• Strong correlation - funding matters
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Insights:

- Increasing budget share alone does not guarantee better outcomes
- Efficiency depends on how funds are utilized, not just allocated

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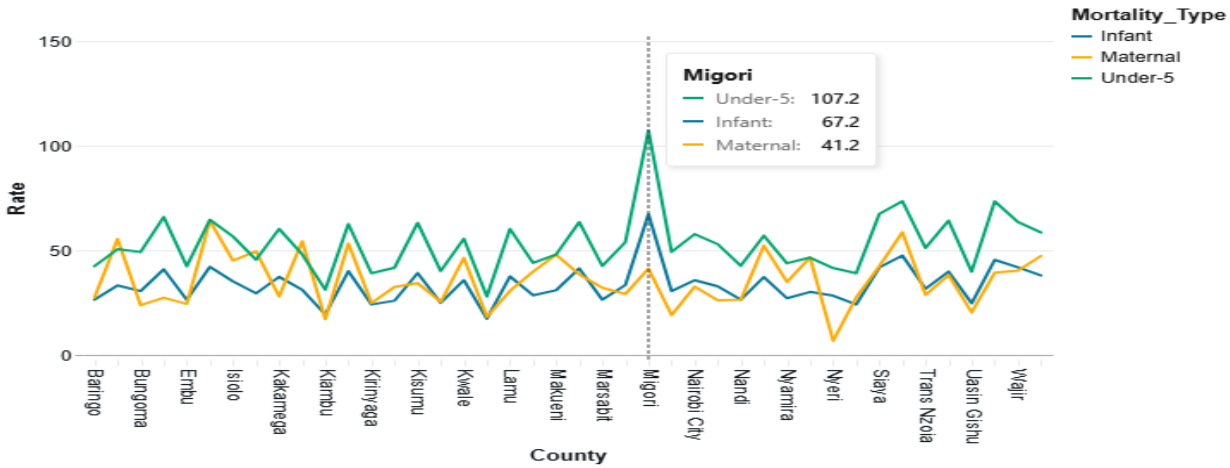
4.6 Mortality Drivers of Poor Performance

Mortality Rates by County



Insights:

- Some counties perform poorly due to high maternal mortality
- Others are driven by high infant or under-5 mortality
- Different counties require different intervention strategies

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  ○  High maternal mortality - poor maternal care systems
  ○  High infant mortality - neonatal care gaps
  ○  High under-5 - broader system issues
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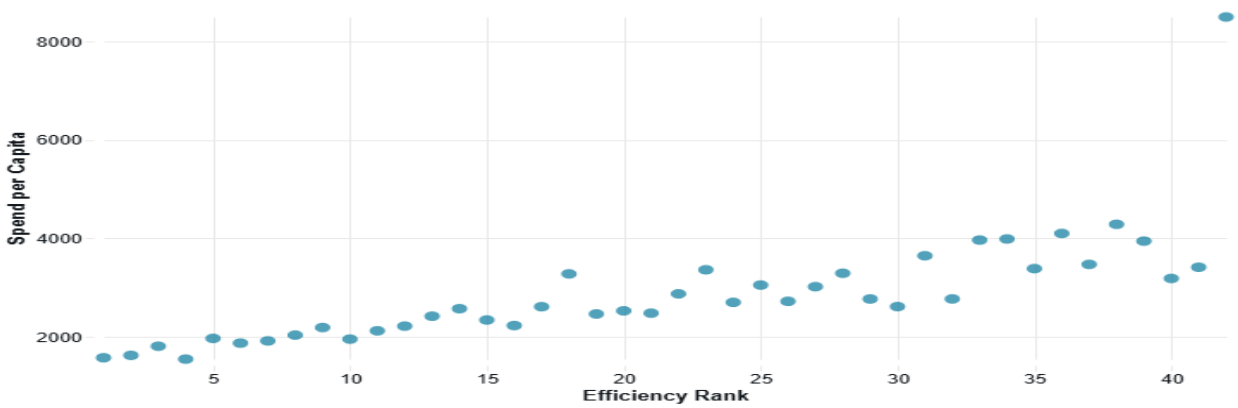
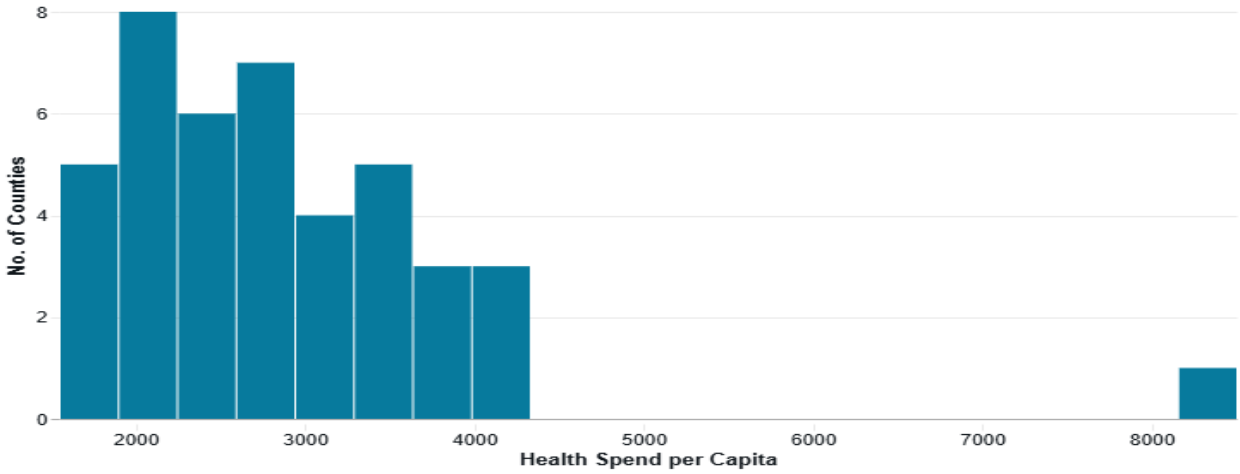
4.7 Spending Distribution

Rank vs Spending



Health Spend per Capita

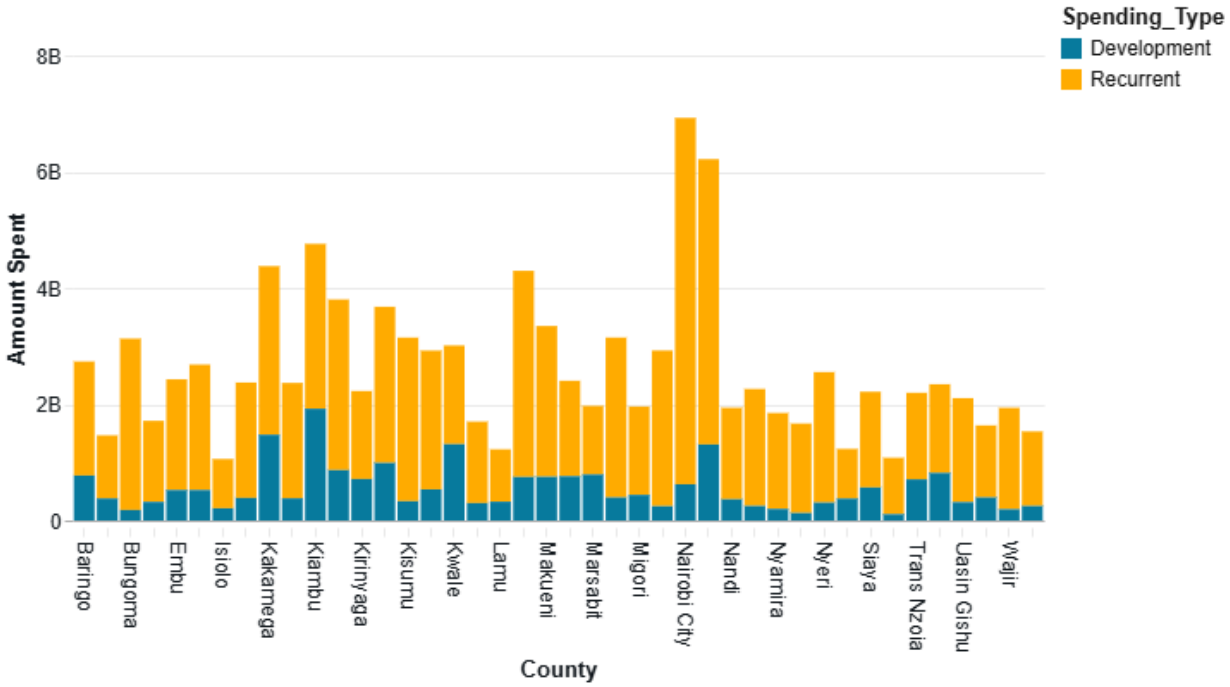


Insights:

- Significant variation in spending across counties
- Indicates inequality in resource allocation

4.8 Spending Composition (Recurrent vs Development)

Stacked Bar Chart



Insights:

- Most counties allocate heavily toward recurrent expenditure
- Limited development spending may constrain long-term improvements

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5. Key Insights

From the analysis, several important patterns emerge:

1. Spending Alone Is Not Enough

Higher expenditure does not consistently lead to better health outcomes.

2. Efficiency Varies Widely

Counties differ significantly in how effectively they utilize resources.

3. Best Practices Exist

High-efficiency counties demonstrate that better outcomes are achievable even with limited resources.

4. Structural Issues Matter

Inefficiency is likely driven by:

- Poor allocation within health budgets
- Weak service delivery systems
- Resource mismanagement

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6. Policy Recommendations

Based on the findings, the following actions are recommended:

6.1 Reallocate Resources Strategically

- Shift funding toward high-impact interventions
- Prioritize outcome-driven budgeting

6.2 Benchmark Against High-Performing Counties

- Identify and replicate best practices
- Encourage knowledge sharing across counties

6.3 Improve Spending Efficiency

- Focus on how funds are used, not just how much is spent
- Strengthen monitoring and evaluation systems

6.4 Optimize Budget Structure

- Balance recurrent and development expenditure
- Invest more in long-term healthcare infrastructure

6.5 Targeted Interventions

- Tailor policies based on specific mortality drivers per county
- Focus interventions where outcomes are weakest

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7. Conclusion

This study demonstrates that efficiency in public health spending varies significantly across counties in Kenya. While some counties effectively translate resources into improved health outcomes, others fall short despite higher levels of spending.

Improving efficiency requires not just increased funding, but smarter allocation, stronger systems, and adoption of proven strategies. By leveraging data-driven insights, policymakers can make more informed decisions to enhance healthcare outcomes nationwide.